Sachin Shankar

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OBJECTIVE

Passionate aerospace and rocketry enthusiast leveraging software development to push the boundaries of propulsion and flight systems. Seeking research, internship, or mentorship opportunities in aerospace engineering and mission-critical software, with long-term goals of contributing to next-generation launch vehicles at organizations like SpaceX or Boeing.

EDUCATION

Tesla STEM High School — Redmond, WA
9th Grade Student — STEM Focus — 2025 – Present

- Rigorous STEM curriculum with deep emphasis on science, technology, engineering, and mathematics.
- Active competitor in Science Olympiad, Science Bowl, and STEM academic competitions.

AWARDS & ACHIEVEMENTS

★ 1st Place — Washington State Science Olympiad (WASO) 2026 — 2026

★ 8th Place — National Science Olympiad 2026 — 2026

★ 2nd Place — National Science Bowl (NSB) Regional Championship 2026 — 2026

★ 21-Time Medalist — Science Olympiad (Multiple Events & Seasons) — 2023 – 2026

- Sustained multi-season excellence across Science Olympiad events spanning physics, chemistry, engineering design, and earth science.

PROJECTS

SuperWin — Windows Multi-Tool Utility — Ongoing
Solo Developer — C++ / WinUI 3 / C++/WinRT

- Built a native Windows desktop application featuring 20 tools across six categories: multi-level calculators with a full CAS and graphing engine (TI-Nspire CX II-style, MathLive/WebView2), a Python IDE, file converter, real-time system diagnostics with always-on-top mini-monitor, process-wide clipboard manager with global hotkey, rich-text editor, JSON formatter, GUID generator, WASAPI volume control, and more.
- Architected as a single `SuperWin.exe` process using WinUI 3 code-first (no XAML markup), C++20, Win32, CMake + Ninja. Decoupled `superwin_core` static library keeps all testable logic outside the UI; ships with Catch2 unit tests.
- Self-contained deployment (Windows App Runtime bundled), in-place automatic updater via WinHTTP appcast, per-user install with no UAC. Publicly released at shankar-sachin.github.io/superwin.

Space Odyssey — Discord Game Bot — Ongoing
Solo Developer — JavaScript

- Designed and built a space-themed Discord game bot in JavaScript where players construct and manage space stations, compete for resources, and progress through in-game events.
- Integrated third-party image generation APIs to produce dynamic in-game visuals; built custom server workflows to automate game state management and event handling.

Solid Rocket Fuel Formulation & Engineering — Ongoing
Independent Research Project

- Independently researching novel solid rocket propellant compositions targeting improved specific impulse (I_{sp}), burn stability, and cost per kg versus conventional formulations (e.g., APCP).
- Using OpenRocket for flight simulation and trajectory modeling; vehicle geometry modeled in CAD software. Project being developed toward WSSEF/ISEF submission.

TECHNICAL SKILLS

Programming	Python (Upper-Intermediate), JavaScript (Upper-Intermediate), HTML/CSS (Upper-Intermediate), Java (Intermediate), C/C++ (Novice — Actively Learning)
Tools	OpenRocket, Git/GitHub, CAD Software (Novice — Actively Learning)
Mathematics	Calculus AB (Differential & Integral Calculus), Introductory Series & Sequences
Domains	Rocketry & Propulsion, Aerospace Systems, Software Engineering, Automation

EXTRACURRICULARS & INTERESTS

- **Science Olympiad** — Multi-season competitor with 21 medals; events spanning physics, chemistry, engineering design, and earth science.
- **Rocketry** — Hands-on interest in solid propellant design, rocket motor engineering, and simulation-driven flight planning via OpenRocket.
- **Aerospace Engineering** — Passionate about applying software to accelerate propulsion research, avionics development, and launch vehicle design.
- **Competitive STEM** — Consistent top performer at regional and national levels across science bowls and olympiads.

CAREER GOALS

Aspiring to pursue aerospace engineering at the university level and contribute to the commercial spaceflight industry. Particularly interested in propulsion systems, avionics software, and launch vehicle design — with SpaceX and Boeing as long-term target employers.